

Workbook 2: Tinker Level

1. Functions & Overloading

Ex 1.1: Write three overloads of `process`: one for `int`, `float`, and `double`.
Tinker: Call `process(1.5)`. Does it call the `float` or `double` version? Why?

Ex 1.2: Try to overload a function by changing **only** the return type.
Does the compiler allow it? Document the error.

2. Templates & Specialization

Ex 2.1: Create a template `is_small<T>` that returns `true` if `sizeof(T) <= 4`.
Specialization: Make it return `false` specifically for `int` even though it is usually 4 bytes.

Ex 2.2: Create a template that takes two different types `T` and `U` and returns their sum. Tinker: What happens if you sum a `char` and an `int`?

3. Lambdas & Requires

Ex 3.1: Write a lambda that captures a local variable by reference. Modify the variable inside the lambda. Check the variable's value outside. Tinker: Now capture it by value and try to modify it—what keyword do you need?

Ex 3.2: Write a template with `requires std::is_integral_v<T>`. Tinker: Try passing a `bool`. Is a boolean considered an integral type in C++?

4. Function Pointers

Ex 4.1: Create a pointer to a function that takes two doubles. Assign a lambda to it. Tinker: Can you assign a lambda that has a **capture** to a raw function pointer?

5. Fold Expressions & constexpr

Ex 5.1: Use a **unary left fold** to subtract a pack of numbers from 100. Tinker: Does the order of the pack change the result compared to a right fold?

Ex 5.2: Write a `constexpr` function. Try to put a `std::cout` inside it. Does it still compile? Why or why not?